Privacy-Preserving Federated Multimodal Latent Synthesis for Chest X-Ray and ECG Diagnosis
A Completed Implementation Report

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Abstract

In our study, we developed a privacy-preserving federated multimodal learning pipeline using open-source chest X-ray and ECG datasets. Our implemented system combines a chest X-ray branch and an ECG branch, learns a shared latent space under federated training, exports protected client-side latent summaries, aggregates them on the server, trains a class-conditional latent diffusion model, generates synthetic data in the shared latent space, and trains a downstream classifier on that synthetic data. We then evaluate the learned classifier on real test data and compare it against a real-latent ceiling baseline. Our study also includes a calibration probe and a sequence of post hoc failure analyses that identify the dominant residual bottleneck in the synthetic branch. The completed work is methodologically strong, but its scope must be stated precisely: metadata integration was not activated, and true patient-level pairing across modalities was not available in the chosen open-source datasets. Therefore, our finished system is best described as a federated, privacy-preserving, class-aligned multimodal latent learning pipeline rather than a fully patient-aligned clinical multimodal cohort study.

1 Our Study

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2 Background of the Study

Modern clinical machine learning is increasingly multimodal. In practice, clinically useful decision systems may require chest X-rays, ECG signals, and structured patient information. However, these data are usually distributed across hospitals and stored under strict privacy constraints. Classical centralized learning assumes that all records can be pooled at one site, but that assumption is often unrealistic in medicine [6, 9, 8]. At the same time, multimodal learning systems are often sensitive to missing modalities and to imperfect alignment between different data sources [2]. These challenges motivate a design in which the model can learn from multiple modalities, tolerate incomplete modality availability, and avoid direct sharing of raw patient data.

Diffusion models and latent generative models have recently become attractive because they can synthesize useful training data in a learned representation space rather than in raw input space [3]. In medical settings, this idea is especially appealing because latent-space synthesis may reduce direct exposure of raw images and signals while still supporting downstream tasks. Nevertheless,

privacy protection cannot be assumed merely because the data are transformed. Practical systems still require clipping, noise injection, restricted server-side aggregation, and utility auditing.

For our study, we used lightweight open-source data to complete the full engineering pipeline. The chest X-ray branch was built on PneumoniaMNIST from MedMNIST v2, while the ECG branch was built on the MIT-BIH Arrhythmia Database [10, 7]. ChestMNIST was loaded for inventory and secondary context, but it was not used as the primary image training branch in the completed multimodal pipeline. The resulting system provides a validated proof of method for privacy-preserving multimodal latent synthesis with open-source data, while also revealing the limitations that remain before a fully patient-aligned deployment can be claimed.

3 Previous Limitations in This Research Area

Previous work in the broader area of multimodal medical machine learning and federated learning has several recurring limitations.

Limitation	**Why it matters**	**Reference**
Centralized data pooling	Many high-performing multimodal systems assume all data can be collected centrally, which is often infeasible for protected hospital data.	[6, 9, 8]
Missing-modality fragility	Real clinical workflows rarely provide every modality for every patient, yet many multimodal models are designed for fully observed inputs.	[2]
Weak privacy accounting in high-dimensional learning	Privacy mechanisms can substantially reduce utility when applied naively, especially in deep models and high-dimensional representations.	[1]
Limited use of synthetic latent training	Many medical systems train directly on real data and do not test whether protected latent summaries can support synthetic downstream training.	[3]
Dataset fragmentation	Public image, ECG, and metadata sources often exist separately, making true patient-level alignment difficult in open research environments.	[5, 4, 7, 10]

In short, previous studies often solved only part of the problem: either multimodal prediction, or federation, or privacy, or synthesis, but not the full chain in a single executable workflow. Our work was designed to address that broader chain in one system.

4 Objectives of This Work

The original objective of our study was to develop a privacy-preserving federated multimodal pipeline that can combine chest X-ray, ECG, and metadata for downstream disease prediction using synthetic data generation in a shared latent space.

The implemented objectives that were completed in the notebook are the following:

1. Prepare a multimodal workspace using a chest X-ray branch and an ECG branch.

2. Build modality-specific encoders.

3. Introduce missing-modality handling through a modality mask.
4. Fuse modality embeddings into a shared representation.
5. Train the shared representation under federated averaging.
6. Export private client-side latent summaries.
7. Aggregate protected latent summaries on the server.
8. Train a latent diffusion model on aggregated summaries and replay latents.
9. Generate synthetic multimodal latent samples.
10. Train a downstream classifier only on the synthetic latent data.
11. Compare the synthetic branch against a real-latent ceiling baseline.
12. Diagnose the residual performance gap using calibration, ranking, geometry, and score-collapse analyses.

The following originally intended objectives were *not* completed in the current implementation:

1. Real metadata integration, because no aligned metadata table was activated in the finished notebook and the metadata branch remained disabled.
2. True patient-level pairing between chest X-ray and ECG, because the public datasets used for the completed system were not jointly indexed by patient.
3. Full formal Step 13 evaluation, including fairness, uncertainty, and external validation.

5 Approach Considered to Fulfill the Objectives

We adopted a staged approach.

1. Use open-source data to make the multimodal pipeline executable end to end.
2. Harmonize the image and ECG branches into a common binary label space.
3. Train a federated multimodal fusion model with explicit modality masking.
4. Export shared latents at the client side rather than raw data.
5. Summarize each client distribution with clipped and noised per-class Gaussian mixture models.
6. Aggregate the protected summaries server-side.
7. Train a class-conditional latent diffusion model using aggregated synthetic latents plus replayed real latents.
8. Generate synthetic shared-latent samples.
9. Train the final disease classifier only on synthetic latent samples.
10. Measure the remaining gap against a classifier trained on real latent samples.

This design directly targeted privacy, multimodality, missing modalities, and synthetic downstream training in one integrated pipeline.

6 Materials and Completed Scope

6.1 Datasets Used in the Completed System

Dataset	Role	Use in the completed pipeline	Reference
PneumoniaMNIST	Primary chest X-ray branch	Used as the main image modality for training, validation, testing, encoder learning, and multimodal pairing.	[10]
ChestMNIST	Secondary chest inventory	Loaded and inventoried during setup, but not used as the primary image branch in the completed fusion pipeline.	[10]
MIT-BIH Arrhythmia Database	ECG branch	Used to build beat-level ECG samples with binary normal/abnormal targets.	[7]
Metadata branch	Planned only	Declared in configuration, but disabled in the finished implementation with `USE_METADATA=False`.	—

6.2 Important Scope Clarification

The finished system is multimodal at the level of model architecture and latent fusion, but not at the level of patient-registered hospital records. Because the image and ECG sources were not truly patient-aligned, we constructed class-aligned multimodal pairs rather than true patient pairs. This decision allowed our study to be completed and audited with open-source data, but it also defines the boundary of the current claims.

7 Step-by-Step Description of the Completed Workflow

7.1 Step 1: Data acquisition and environment setup

The notebook first installed the required packages and copied the available datasets into a working directory. It inventoried PneumoniaMNIST, ChestMNIST, and MIT-BIH records. This step verified that the environment had the image and ECG sources necessary for the pipeline.

7.2 Step 2: Shared binary target design

The image branch used PneumoniaMNIST as the primary chest X-ray task, with labels mapped directly to a binary target. The ECG branch used the MIT-BIH annotation symbols. A binary mapping was defined such that standard normal beat symbols were mapped to class 0 and abnormal beat symbols were mapped to class 1. This yielded a common label space:

```
0 = negative_or_normal,   1 = positive_or_abnormal
```

This label harmonization was essential because the multimodal pipeline needed a shared target even though the modalities came from different public sources.

7.3 Step 3: ECG preprocessing and record-level split

The ECG branch was built from MIT-BIH waveform records. The records were split at the record level into training, validation, and test partitions using a 70/15/15 strategy. For each record, we selected the MLII channel when available, otherwise the first available lead. Beats were extracted around the annotated R-peak using a fixed window of 256 samples. Each beat was mean-centered and standardized before storage. This produced beat-level ECG tensors with record-level train/validation/test separation.

7.4 Step 4: Federated client construction

The image and ECG training sets were each partitioned across five clients using constrained Dirichlet splitting with $\alpha = 0.5$. The split function enforced minimum sample counts, minimum per-class representation, and an upper bound on majority-class dominance. This produced a stable non-IID training environment for both modalities.

7.5 Step 5: Modality-specific dataset preparation

Separate dataset and dataloader classes were created for images and ECG beats. Images were normalized by the training mean and standard deviation after conversion to the $[0, 1]$ range. ECG beats were normalized using the training-set mean and standard deviation. This step created the local dataloaders that each client used during federated training.

7.6 Step 6: Modality encoders and gated late fusion

The image encoder was a lightweight convolutional network built from repeated 2D convolution, batch normalization, and GELU blocks, followed by adaptive average pooling and a layer-normalized projection head. The ECG encoder used an analogous 1D convolutional design, again ending with adaptive pooling and a layer-normalized projection head.

Fusion was performed with a gated attention block. The image embedding and ECG embedding were first projected into a common fusion space. They were then treated as tokens in a multi-head attention layer. A learned gate blended the attended tokens with the original tokens. Finally, the fused representation was masked and averaged according to modality availability. This design allowed the model to operate with both modalities or with only one modality.

7.7 Step 7: Class-aligned multimodal pairing and missing-modality masking

Because the image and ECG sources were not patient-aligned, we formed class-aligned multimodal pairs. For each client and each class, image examples and ECG examples from the same binary class were paired, with ECG sampling performed with replacement when the counts were unequal. The training dataloader then applied modality dropout with probability 0.25, and a modality mask ensured that at least one modality remained present in every sample. This simulated the missing-modality regime that is common in real clinical workflows.

7.8 Step 8: Federated multimodal training

The global multimodal fusion model was trained with federated averaging. The implemented schedule used eight communication rounds and four local epochs per round. Each client received the current global model, optimized it locally with AdamW and cross-entropy loss, and returned its updated parameters. The server aggregated the client parameters with a sample-size-weighted

average. Validation statistics were tracked after each round, and the best global model was selected by validation AUC with loss as a tie-breaker.

7.9 Step 9: Latent export and client-side private summarization

After federated training, each client's shared latent representations were exported from the best global model. For every client, the latent vectors were debiased by subtracting the client mean. A clipping norm was then estimated from the pooled latent norm distribution using the 97.5th percentile. For each class within each client, a diagonal-covariance Gaussian mixture model was fitted to the clipped latent vectors, subject to a minimum sample requirement.

Privacy protection was applied directly to the latent summaries: the component means and covariances were perturbed with Gaussian noise scaled by the clipping norm and a privacy parameter $\epsilon = 0.5$. In the completed implementation, each client payload therefore contained only a bias vector, class-wise private GMM parameters, class sample counts, and average modality availability, rather than any raw image or ECG samples.

7.10 Step 10: Server-side aggregation of protected payloads

On the server, the private class-wise latent GMMs from all clients were aligned and aggregated. Component matching used the Hungarian assignment algorithm when available, with a greedy fallback otherwise. After matching, the server computed weighted averages of component means, covariances, and mixture weights. The latent bias vectors and average modality masks were also aggregated. This yielded a global protected latent summary for each class.

7.11 Step 11: Multimodal latent diffusion training set

The diffusion training set combined two sources:

1. synthetic latent samples drawn from the aggregated class-wise GMM payloads, and
2. replay latent vectors exported from the trained global model.

The combined training set was class-balanced with a weighted random sampler. Validation latents were extracted from the real multimodal validation split and used to monitor denoiser performance.

7.12 Step 12: Class-conditional latent diffusion model

The generative model was a class-conditional latent denoiser with sinusoidal time embeddings, class embeddings, residual MLP blocks, and an exponential moving average copy for stable generation. The denoiser was trained for up to 250 epochs with 40 diffusion steps, cosine learning-rate decay, classifier-free guidance, and early stopping. The model operated entirely in the shared latent space rather than in raw image or signal space.

7.13 Step 13: Synthetic latent generation

After denoiser training, the EMA model was used to sample 2,500 synthetic shared-latent vectors per class. These synthetic samples formed the training data for the downstream classifier. Validation diagnostics compared synthetic and real latent distributions by class using mean gap, covariance gap, and radial-basis-function MMD.

7.14 Step 14: Synthetic-only downstream classifier

The downstream classifier was a small MLP operating on the synthetic latent vectors. It used AdamW optimization, validation-based early stopping, and a threshold selected from the validation split to maximize balanced accuracy and F1 score. The classifier was trained on synthetic data and then evaluated on real test data extracted from the trained multimodal model.

7.15 Step 15: Real-latent ceiling baseline

To establish an upper bound, we extracted real multimodal train latents from the actual training sources and trained the same classifier family directly on those real latents. This provided a fair real-data ceiling baseline against which the synthetic-data branch could be judged.

7.16 Step 16: Final comparison and performance diagnosis

The notebook compared the synthetic-data branch against the real-data ceiling on real test data across threshold, validation AUC, test AUC, accuracy, precision, recall, specificity, F1 score, and balanced accuracy. It then ran a targeted sequence of diagnostics to understand the residual gap.

8 Main Findings of the Completed System

The completed findings are summarized below. Only values that were verified in the preserved notebook execution trace are listed numerically.

Finding	Value		Interpretation	Source
Synthetic threshold after calibration	0.6200 0.8284	→	The synthetic branch benefited materially from post hoc threshold correction.	Notebook calibration probe
Specificity after calibration	0.6667 0.7265	→	The most obvious false-positive burden was reduced through threshold calibration.	Notebook calibration probe
Recall after calibration	0.9974 0.9923	→	Calibration improved specificity while preserving very high sensitivity.	Notebook calibration probe
Precision after calibration	0.8330 0.8581	→	Precision improved once the threshold was tightened.	Notebook calibration probe
Balanced accuracy after calibration	0.8321 0.8594	→	Calibration materially improved overall operating-point quality.	Notebook calibration probe

Ambiguous-negative logit gap	+2.1626	In ambiguous negative regions, the synthetic branch assigned substantially more positive evidence than the real-latent ceiling.	Notebook logit-collapse audit
Confident-positive logit gap	+5.7734	In confident positive regions, the real-latent ceiling produced much stronger positive margins than the synthetic branch.	Notebook logit-collapse audit
Residual utility gap	qualitative but persistent	Calibration improved the operating point, but ranking quality and latent class structure still lagged the real-latent ceiling.	Final comparison and audits

The key scientific conclusion of our study is therefore clear: calibration alone was not sufficient. It improved specificity and balanced accuracy, but the dominant remaining problem was weaker latent ranking structure in the synthetic branch.

9 Detailed Interpretation of the Diagnostics

The post hoc diagnostics showed a consistent pattern.

1. The false-positive audit showed that the negative-class operating point was the first visible weakness.
2. The threshold probe showed that this weakness was not purely a calibration failure, because calibration improved the operating point but did not remove the ranking gap.
3. The ranking-gap audit showed that the synthetic probabilities separated positive and negative samples less effectively than the real-latent ceiling.
4. The hard-negative geometry audit showed that many difficult false alarms occurred in latent neighborhoods that were locally ambiguous.
5. The score-compression audit showed that the synthetic branch had weaker dynamic range across evidence bins than the real-latent ceiling.
6. The tail-collapse audit showed an asymmetry: ambiguous negatives were inflated upward, while confident positives were compressed into a narrow ceiling region.
7. The logit-scale audit confirmed that this problem was more severe in raw decision-margin space than in ordinary probability space.

This set of results supports a coherent explanation: the latent diffusion model produced useful synthetic training data, but it did not preserve class geometry sharply enough. The downstream classifier therefore learned a decision function with reduced ranking power, especially in regions that should have been strongly negative or strongly positive.

10 Methods That Failed, Were Repaired, or Were Explicitly Ignored

The completed system included several method-level failures or rejected directions. These are important to document.

10.1 Metadata branch was declared but not used

The notebook reserved a metadata embedding size and a metadata flag, but the completed implementation kept `USE_METADATA=False`. This branch was not activated because a real aligned metadata table was not available in the current open-data environment. We therefore did not claim completed metadata fusion.

10.2 True patient-level cross-modal pairing was not available

The open datasets used in the completed system did not provide patient-registered chest X-ray and ECG samples from the same individuals. Because of this, we could not perform true patient-level multimodal fusion. We replaced that step with class-aligned multimodal pairing. This was a deliberate engineering compromise, not a hidden assumption.

10.3 Calibration-only validation-label recovery failed

At one stage, we attempted a direct calibration-only retuning procedure that depended on recovering validation labels from runtime artifacts. This repeatedly failed with unresolved validation-label recovery. We therefore replaced it with a narrower threshold probe that relied only on already verified runtime objects. This narrower probe succeeded and yielded the calibrated operating-point results listed above.

10.4 Early false-positive audit variants failed due to unstable runtime naming

Initial false-positive audit versions failed because the expected cached probability variables or classifier names were not reliably present in the runtime. This problem was solved by using auto-discovery anchored to validated summary metrics and by resolving probability sources programmatically.

10.5 The first score-compression audit failed due to a pandas categorical groupby bug

The first implementation of the score-compression audit produced a categorical groupby shape mismatch. That failure was repaired by grouping with `observed=True` and resetting the index explicitly. The repaired audit was then retained and used in the final analysis.

10.6 Additional diagnostics were proposed but not retained unless they supported the active workflow

A margin-response slope audit was proposed after the main collapse analyses. It was not executed because the working rule for the project became stricter: diagnostics were retained only if they produced a direct correction or a necessary explanation for the current workflow. Since the existing audits had already localized the bottleneck, the extra diagnostic was not pursued.

11 What Was Completed Successfully

The following parts of the intended study were completed successfully in executable form.

1. Multimodal training pipeline with image and ECG branches.
2. Missing-modality handling through a modality mask.
3. Federated multimodal training with non-IID client partitions.
4. Private client-side latent summarization with clipping and noise.
5. Server-side aggregation of protected latent payloads.
6. Class-conditional latent diffusion training.
7. Synthetic latent generation.
8. Synthetic-only downstream classifier training.
9. Real-latent ceiling baseline.
10. Utility comparison between synthetic and real-latent branches.
11. Calibration probe and calibrated comparison.
12. Ranking-gap, geometry, score-compression, tail-collapse, and logit-collapse diagnostics.

12 What Was Not Completed

The following items remain outside the completed scope of the current notebook.

1. Real metadata integration.
2. True patient-level chest X-ray and ECG alignment.
3. Formal privacy audit with explicit (ϵ, δ) reporting and leakage testing.
4. Fairness analysis by subgroup.
5. Uncertainty analysis.
6. External validation on a separate hospital or domain.

13 Conclusion

Our study completed an end-to-end privacy-preserving federated multimodal latent synthesis pipeline using a chest X-ray branch and an ECG branch from open-source datasets. The system successfully learned a shared multimodal latent space, exported protected client-side summaries, aggregated those summaries on the server, trained a latent diffusion model, generated synthetic data in latent space, and trained a downstream classifier on that synthetic data. We then evaluated the model on real test data. The comparison against a real-data ceiling baseline showed that the methodology worked, but the synthetic branch remained measurably below the ceiling.

The post hoc analyses led to a precise conclusion. The dominant residual problem was not a simple threshold choice. Calibration improved specificity and balanced accuracy, but the synthetic branch still exhibited compressed score dynamics and weaker ranking structure than the real-data baseline. In particular, ambiguous negatives were pushed too far upward and confident positives were not separated strongly enough in raw logit space. These findings identify our work as a valid completed methodological study with clearly documented boundaries.

14 Next Keywords

Patient-level pairing; metadata integration; formal differential privacy accounting; fairness; uncertainty; external validation; heterogeneous modality federation; real clinical multimodal cohort.

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